



School of Computer Science & IT Devi Ahilya Vishwavidyalaya

SYLLABUS

M. Tech (Information Architecture & Software Engineering) 2 years

Program Educational Objectives (PEOs)

- PEO 1:** Develop software engineering skills for architecting quality products and carve out successful careers in IT industry and academia.
- PEO 2:** Demonstrate ability to manage software projects and participate effectively in group tasks and initiate entrepreneurship activities.
- PEO 3:** Deliver quality software products and perform research in the domain of software engineering.

Program Specific Outcomes (PSOs)

- PSO 1:** Acquire expertise to build real-life software applications using agile approach.
- PSO 2:** Design, implement and integrate software with maintainability, reusability, refactoring and design pattern issues in mind.

I - SEMESTER

CS-4407: INFORMATION SYSTEM DESIGN

Course Outcomes (COs):

- CO1:** Explain the concepts, importance, and elements of designing an information system.
- CO2:** Comparative study of various process models used in information system design.
- CO3:** Demonstrate project planning and management activities in order to manage and estimate software projects.
- CO4:** Discuss structured and object-oriented analysis & design methodologies.
- CO5:** Practice object-oriented modeling using UML for the design of real-life problems.
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Course Contents

UNIT-I

No. of Hours: 8

Introduction to Information System, Challenges in ISD, Elements of ISD, Roles and responsibilities in IS Design, Case Study of Information System Design; Software processes, Information system design models, A comparative study of Information System design models (Traditional, CBD, Agile and RUP process).

UNIT-II

No. of Hours: 8

Project Management and Planning: Project management essentials, Project success and failures, Project Life Cycle, Project team structure and organization. Project planning: Metrics and Measurements, Project Estimation: Effort Estimation Techniques, Staffing and Personnel Planning. Project Scheduling. Software Configuration Management, and Risk Management.

UNIT-III

No. of Hours: 8

Requirements engineering: Requirements Elicitation, Requirements analysis: Structured Vs. Object-oriented analysis, Prototyping analysis; Requirements specification, Requirements validation; Requirements management and its tools. Software Design: Design fundamentals, Design process, Modular design, and Component level design. Structured Vs. Object-oriented design, Refactoring.

UNIT-IV

No. of Hours: 8

Object-oriented analysis and design, UML modeling, Use case diagrams: Use Case scenario and Use Case relationships, Class and Object diagrams, Sequence diagrams, Collaboration diagrams, State-Chart diagrams, Activity diagrams, Component diagrams, Deployment diagrams, Generalizations, Domain Model refinements, Architecture, Packaging model elements. Case Study of large-scale systems.

UNIT-V

No. of Hours: 8

Implementation: Traditional vs. TDD, Testing of information systems, Testing strategies, Levels of testing. Debugging, Automation Testing. Software testing tools.

Text Books:

1. Introduction to Information Systems, O'Brain, Prentice Hall, 1997.
2. Software Engineering: Concepts & Practices- Ugrasen Suman, Cengage Learning Publications, Second Ed. 2022.
3. Object Oriented Analysis and Design Using UML- Ugrasen Suman et al, Cengage Learning, First Edition, 2018.
4. Object Oriented Modeling and Design Using UML-James Rumbaugh, Pearson Education.

Reference Books:

1. Software Engineering-A practitioner's approach- R. S. Pressman, Tata McGraw-Hill International Editions, New York.
2. Object Oriented Analysis and Design with Applications-Grady Booch, Addison Wesley.
3. Object-Oriented Software Engineering: A Use Case Driven Approach- Ivar Jacobson et al, Pearson Education.

Online Resources:

CS-4407: INFORMATION SYSTEM DESIGN - PRACTICAL

Course Outcomes (COs):

CO1: Identify various components of different information systems.

CO2: Design different information systems using concepts of information system design.

CO3: Practice object modeling using UML for the design of real-life problems.

CO4:

CO5:

Course Contents

UNIT-I

No. of Hours: 4

1. Identify the different components in the following systems: Library Information System, Microsoft Office, Cellular Phone.

2. Identify systems that are not information systems?

3. Use the structured design methodology to produce a design for the following:

- A system to convert ASCII to EBCDIC?
- A system that acts as a calculator with only basic arithmetic functions?
- A system that does student registration in the manner it is done at your college?
- A system to manage the inventory at a hardware store?

4. Design an information system for a college to manage student records.

- Identify inputs, processes, and outputs.
- Draw a Context Diagram.
- Draw a Data Flow Diagram (DFD Level 0).
- Design an Entity Relationship (ER) Diagram.
- Create database tables with primary and foreign keys.

UNIT-II

No. of Hours: 4

1. Design an information system for a library.

- Identify system requirements.
- Draw a Context Diagram.
- Draw DFD Level 0 and Level 1.
- Draw an ER Diagram.
- Design tables for Books, Members, Borrowing, and Staff.

2. Create a Use Case Diagram using a UML tool to

- Create a system (e.g., Library Management System).
- Add actors (Student, Librarian).
- Add use cases (Login, Issue Book, Return Book).
- Connect actors with use cases using associations.

3. Draw a Class Diagram using a UML tool:

- Create classes such as Student, Book, Library.

- Add attributes and methods.
- Show relationships:
 - Association
 - Aggregation
 - Composition
 - Inheritance
- 4. Create a Sequence Diagram using a UML tool:
 - Model the "Book Issue" process.
 - Add objects.
 - Draw messages between objects.
 - Show activation bars and return messages.

UNIT-III

No. of Hours: 4

1. Create a Collaboration (Communication) Diagram using a UML tool:

- Represent interactions between objects.
- Number the messages in execution order.

2. Draw an Activity Diagram using a UML tool:

Model an ATM withdrawal or online shopping process. It should include:

- Initial node
- Decision node
- Merge node
- Final node

UNIT-IV

No. of Hours: 4

1. Draw a State Chart Diagram for modeling the states of an ATM card in order using a UML tool.

2. Create a Component Diagram using a UML tool, for representing software modules and dependencies between different components.

3. Create a Deployment Diagram using a UML tool, showing the client computer, Application server, Database server, and connect nodes with communication links.

UNIT-V

No. of Hours: 4

1. Import Java or C++ code into Rational Rose and generate the class diagram.

2. Create a class diagram and generate Java/C++ source code.

3. Create a new Rational Rose project:

- Save the model.
- Add a new diagram.

Text Books:

1. Introduction to Information Systems, O'Brain, Prentice Hall, 1997.
2. Software Engineering: Concepts & Practices- Ugrasen Suman, Cengage Learning Publications, Second Ed. 2022.
3. Object Oriented Analysis and Design Using UML- Ugrasen Suman et al, Cengage Learning, First Edition, 2018.

4. Object Oriented Modeling and Design Using UML-James Rumbaugh, Pearson Education.

Reference Books:

1. Software Engineering-A practitioner's approach- R. S. Pressman, Tata McGraw-Hill International Editions, New York.
2. Object Oriented Analysis and Design with Applications-Grady Booch, Addison Wesley.
3. Object-Oriented Software Engineering: A Use Case Driven Approach- Ivar Jacobson et al, Pearson Education.

Online Resources:

CS-5225: INTERNET PROGRAMMING USING JAVA

Course Outcomes (COs):

CO1: Understanding of OOP concepts clearly.

CO2: Understanding of client-server architecture and its working.

CO3: Understanding of database creation and server technologies.

CO4: Providing knowledge of advanced Java concepts like servlet, JSP, etc.

CO5: Understanding of MVC Architecture.

Course Contents

UNIT-I

No. of Hours: 8

Review of java concepts: Features of Java, Object-oriented programming overview, Introduction of Java Technologies, How to write simple Java programs, Data Types, Variables, Memory concepts, control statements, looping, Method CallStack and Activation Record, Argument Promotion and Casting, Scope of declaration and Method Overloading, String Handling: The String constructors, String operators, Character Exaction, String comparison, String Buffer. Arrays: Declaring and Creating Arrays, Enhanced for Statement, Passing Arrays to Method, Multidimensional Arrays, Variable-Length Argument lists, Using Command-line Arguments.

UNIT-II

No. of Hours: 8

Inheritance: Extending classes & related things, Packages and Interfaces: Defining a package, Understanding CLASSPATH, Access Protection, Importing packages, creating own packages Exception Handling: Introduction, over view of doing it and keywords used, when to use it, Multithreading: What are threads, The java Thread model, Thread priorities, Thread life cycle, Thread Synchronization, Applets: Applet basics, Applet Architecture, Applet life cycle methods, Database connectivity: JDBC, The design of JDBC, Typical uses of JDBC.

UNIT-III

No. of Hours: 8

Introduction to HTTP, web servers and application Servers, Installation of Application servers, Config files, Web.xml. Java Servlet, Servlet Development Process, Deployment Descriptors, Generic Servlet, Lifecycle of Servlet. Servlet Packages, Classes, Interfaces, and methods; handling forms with a servlet; various methods of Session Handling; various elements of deployment descriptors.

UNIT-IV

No. of Hours: 8

JSP Basics: JSP lifecycle, Directives, scripting elements, standard actions, implicit objects. Connection of JSP and Servlet with different databases viz. Oracle, MS-SQL Server, MySQL.java.sql Package. Querying database, adding records, deleting records, modifying records, types of Statement. Separating Business Logic and Presentation Logic, Building and using a JavaBean. Session handling in JSP, Types of errors and exception handling.

UNIT-V**No. of Hours: 8**

MVC Architecture: Introduction to Remote Method Invocation, Introduction to Enterprise Java Bean, Types of EJB, Creating and working with Session Bean.

Text Books:

1. Java2:The Complete Reference by Herbert Schildt, Tata McGraw-Hill, 8th Edition, 2011.
2. K. Mukhar, "Beginning Java EE5: From Novice to Professional", Wrox Press.

Reference Books:

1. The Java Programming Language, Ken Arnold, James Gosling, David Holmes, 3rd Edition, Pearson Education, 2000.
2. Head First Java, Kathy Sierra, Bert Bates, O'Reilly Publication, 2nd Edition, 2005

Online Resources:

1. NPTEL Course Video Lectures on "Programming in Java", by Prof. Debasis Samant, IIT Kharagpur
2. Swayam Course Video Lectures on "Web Technology", By Prof. Dr. Ashutosh Kumar Bhatt

CS-5225: INTERNET PROGRAMMING USING JAVA - PRACTICAL

Course Outcomes (COs):

CO1: Develop and understand Java programming concepts including inheritance, polymorphism, interfaces, packages, multithreading and exception handling.

CO2: Apply the above concepts to develop professional and efficient Java code.

CO3: Develop real-world web applications using advanced Java concepts like JDBC, Servlet, JSP, and MVC.

CO4:

CO5:

Course Contents

UNIT-I

No. of Hours: 8

1. Create a program that reads jagged array dimensions from command-line arguments, dynamically creates the array, and demonstrates implicit type promotion during element assignment.
2. Write an overloaded process() method where one version accepts a multi-dimensional array and another accepts varargs. Demonstrate recursion with explicit narrowing type casting and trace the call stack.
3. Design a class to demonstrate variable shadowing using a global String and a local StringBuffer, displaying their hash codes in each loop iteration.
4. Implement a constant-time string comparison using charAt() without using the equals() method.
5. Write a program that creates a deep copy of a 3D array, rotates (or flips) the copy, and returns it without modifying the original array.
6. Create overloaded methods calculate(int... nums) and calculate(double num, int... nums), then invoke them using short, char, and float values to demonstrate method resolution.
7. Write a program to manually sort an array of strings by converting them into byte arrays without using built-in sorting methods.
8. Create a recursive program that tracks method calls, terminates recursion based on a condition, and continues execution iteratively to avoid StackOverflowError.
9. Design a program that demonstrates upcasting and downcasting using an array of objects and updates object states using an enhanced for loop.
10. Write a program that performs repeated string concatenation inside a loop to demonstrate heap allocation and garbage collection through reference reassignment.

UNIT-II

No. of Hours: 8

1. Develop a University Management System using multilevel inheritance, abstract classes, method overriding, and runtime polymorphism.
2. Create a multi-package Banking System demonstrating custom packages, access modifiers, package importing, and CLASSPATH configuration.

3. Design a custom utility package for matrix and statistical operations and execute it using CLASSPATH.
4. Implement an Online Examination System using user-defined exceptions, exception propagation, and nested exception handling.
5. Develop a multithreaded Railway Reservation System demonstrating thread priorities, synchronization, and thread life cycle.
6. Implement the Producer–Consumer problem using multiple threads with wait(), notify(), and synchronization.
7. Create a Java Applet to draw geometric shapes using mouse events and demonstrate the complete applet life cycle.
8. Develop a JDBC-based Student Management System implementing CRUD operations, transactions, and PreparedStatement.
9. Create a multithreaded Banking System that performs synchronized database transactions using JDBC.
10. Develop a Library Management System integrating inheritance, packages, exception handling, multithreading, interfaces, and JDBC.

UNIT-III

No. of Hours: 8

1. Develop a login system using Java Servlets that authenticates users, manages sessions, and restricts access to protected resources using deployment descriptors.
2. Create a multi-module web application with multiple servlets configured entirely through web.xml, demonstrating servlet mapping, initialization parameters, and load-on-startup.
3. Design an online examination system using Servlets that processes dynamic forms, validates user input, maintains session state, and generates a result report.
4. Develop a shopping cart application using Servlets that demonstrates all session tracking techniques (Cookies, URL Rewriting, Hidden Fields, and HttpSession) and compares their behaviour.
5. Implement a file upload and download system using Servlets that validates uploaded files, stores metadata, and handles exceptions gracefully.
6. Create a servlet-based Employee Management System implementing CRUD operations with HTML forms, request dispatching, and configurable servlet initialization parameters.
7. Develop a secure online banking portal using GenericServlet and HttpServlet to demonstrate authentication, authorization, session timeout, and servlet life cycle methods.
8. Build a visitor analytics application that records client requests, browser information, session duration, and page visits using Servlet APIs and deployment descriptor configuration.
9. Create a servlet application that dynamically generates reports based on user input, uses request forwarding and redirection, and manages concurrent client requests efficiently.
10. Develop a complete e-commerce web application using Servlets, deployment descriptors, session management, and servlet life cycle methods, with configurable initialization parameters defined in web.xml.

UNIT-IV

No. of Hours: 8

1. Develop a JSP-based Student Management System that performs CRUD operations using JavaBeans, JDBC, and MySQL while separating business logic from presentation logic.
2. Create a secure JSP login application that authenticates users using a database, manages sessions, and handles invalid login attempts with custom error pages.
3. Design an online banking portal using JSP, JavaBeans, and JDBC to perform account management, fund transfers, and transaction history with proper exception handling.
4. Develop a JSP application demonstrating the complete JSP lifecycle, directives, scripting elements, standard actions, and implicit objects in a single project.
5. Create a database-driven Inventory Management System using JSP and JDBC that supports Statement, PreparedStatement, and CallableStatement, and compares their usage.
6. Implement a JSP-based Employee Management System that connects to Oracle, MySQL, or MS SQL Server and performs CRUD operations using JavaBeans and MVC architecture.
7. Develop a JSP application that uses session management to implement a multi-step online examination system with automatic session timeout and recovery.
8. Create a dynamic reporting system using JSP and JDBC that executes parameterized database queries and displays formatted reports with robust exception handling.
9. Design a JSP-based Library Management System using JavaBeans, JDBC, and MVC architecture, supporting book issue/return operations and database transaction management.
10. Develop a complete JSP-based E-Commerce application integrating JavaBeans, session management, JDBC, MVC architecture, database CRUD operations, and comprehensive error and exception handling.

UNIT-V

No. of Hours: 8

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2. K. Mukhar, "Beginning Java EE 5: From Novice to Professional", Wrox Press.

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2. Head First Java, Kathy Sierra, Bert Bates, O'Reilly Publication, 2nd Edition, 2005.

Online Resources:

1. NPTEL Course Video Lectures on "Programming in Java", By Prof. Debasis Samant, IIT Kharagpur
2. Swayam Course Video Lectures on "Web Technology", By Prof. Dr. Ashutosh Kumar Bhatt.

CS-6314: INFORMATION ARCHITECTURE

Course Outcomes (COs):

- CO1:** Develop conceptual understanding of UX/UI design and information architecture.
- CO2:** Build a theoretical foundation on principles and concepts of designing effective information architecture.
- CO3:** Study existing portals on aspects of organizing, classification, navigation, searching & labeling information and prepare comprehensive reports on strength/weakness.
- CO4:** Understand TOGAF(The Open Group Architecture Framework) for building information architecture documents.
- CO5:** Supplement the fundamentals learned with case studies of modern web infrastructure like Cloud, Security and Mobile.
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Course Contents

UNIT-I

No. of Hours: 8

Introduction to UX/UI: Design Issues, tools & processes, Design Principles of good UX/UI design.

UNIT-II

No. of Hours: 8

Introduction to Information Architecture: Principles of information architecture, role of information architect, areas of information architecture, types of architecture, system architecture, enterprise architecture, application architecture, internet architecture, research and practice in information architecture.

UNIT-III

No. of Hours: 8

Organizing information: Organizing information, organizational challenges, organizing websites and intranets, creating cohesive organization system, organizing WWW, browser navigation features, building context, improving flexibility, types of navigation systems, integrated navigation elements, remote navigation elements, designing elegant navigation systems.

UNIT-IV

No. of Hours: 8

Labeling Systems: Labeling systems, not labels, types of labeling systems, creating effective labeling systems, fine-tuning the labeling system, non-representational labeling systems, double challenge, searching website, website understanding how users search, designing the search interface, reference interview, indexing the right stuff, search or not to search.

UNIT-V

No. of Hours: 8

The Open Group Architecture Framework: Introduction to TOGAF 9, TOGAF 9 management overview, TOGAF 9 components, introduction to ADM, ADM phases, ADM deliverables, enterprise continuum, architecture repository, architecture governance, views and viewpoints, architecture building blocks.

Text Books:

1. Information Architecture on the World Wide Web, First Edition, Peter Morville.

Reference Books:

1. TOGAFVersion9–The Open Group Architecture Framework.Publisher–The Open Group.
2. TOGAF–Quickstart guide for Enterprise Architect, Wolfgang Keller.

Online Resources:

CS-6315: ADVANCED SOFTWARE ENGINEERING

Course Outcomes (COs):

- CO1:** Implement reusability mechanisms for producing application systems.
- CO2:** Explore maintenance and reengineering approaches for legacy systems.
- CO3:** Understand the properties and methods of designing a reliable and safe system.
- CO4:** Analyze and apply CASE in SDLC phases.
- CO5:** Study of advanced concepts in various domains to produce an effective software system.
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Course Contents

UNIT-I

No. of Hours: 8

Introduction to Software Reuse: What is Software Reuse? Reuse types, Reuse Approaches, Reuse Technology, Reuse benefits & barriers, Reuse success & failure Factors, CBSE Process, Reuse Driven Software Engineering is a business.

UNIT-II

No. of Hours: 8

Architectural Concepts in Reuse: Application and component systems, Application families allow significant reuse, Developing Application Systems from Reusable Components: Reuse Variability, Facades for Component System Internals and Externals. Organizing a system in Layered Architecture.

UNIT-III

No. of Hours: 8

Software change, Software Evolution, Software maintenance: Models and Metrics, Reengineering: Reengineering Process and Activities: Program Comprehension, Reverse Engineering, Restructuring, Forward Engineering, Re-documentation. Software Aging.

UNIT-IV

No. of Hours: 8

Software Dependability, Software Safety, Software Availability, Software Reliability: Metrics, Approaches and Models. Software Quality: Quality Factors, Verification & Validation (V&V), SQA. Computer-Aided Software Engineering (CASE): Scope and Technology, CASE support in SDLC, Second Generation CASE Tools, Architecture of a CASE Environment.

UNIT-V

No. of Hours: 8

Usability Engineering: HCI, Types of UI, Component-Based GUI Development, Usability Engineering Process and Methods. Aspect-Oriented Software Engineering. Cleanroom Software Engineering and Crowdsourcing. Artificial Intelligence & Machine Learning in SDLC.

Text Books:

1. Software Reuse Architecture, Process and Organization for Business Success- by Ivar Jacobson, Martin Griss, Patrick Johnson, First Edition, Pearson Education, 2000.

2. Software Engineering: Concepts & Practices- Ugrasen Suman, Cengage Learning Publications, Second Ed. 2022.
3. Software Engineering- Ian Somerville, Pearson Education Asia, 10th Edition, 2016.
4. Software Engineering: A Practitioner's Approach- R. S. Pressman, Tata McGraw-Hill International Editions, New York.

Reference Books:

Online Resources:

CS-6315: ADVANCED SOFTWARE ENGINEERING - PRACTICAL

Course Outcomes (COs):

- CO1:** Implement reusability mechanisms for producing application systems.
- CO2:** Explore maintenance and reengineering approaches for legacy systems.
- CO3:** Understand the properties and methods of designing a reliable and safe system.
- CO4:** Analyze and apply CASE in the SDLC phases.
- CO5:** Study of advanced concepts in various domains to produce an effective software system.
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Course Contents

UNIT-I	No. of Hours: 4
1. Implementation of reusable concepts in Java using reusable libraries. 2. Measurement of cost-effectiveness in reuse-based development. 3. Architectural design of reuse-based development.	
UNIT-II	No. of Hours: 4
1. Implementation of reuse-based modeling. 2. Implementation of use case models and use case relationships in reuse. 3. Implementation of faced and reusability mechanisms. 4. Design of use case components and layered system.	
UNIT-III	No. of Hours: 4
1. Study of legacy systems. 2. Observations on program comprehension: static and dynamic mechanisms. 3. Study of reverse engineering using a case study.	
UNIT-IV	No. of Hours: 4
1. Study of CASE tools. 2. Apply CASE tools in software development. 3. Analysis of software safety for designing a secure system.	
UNIT-V	No. of Hours: 4
1. Analysis of HCI on web-based systems. 2. Implementation of aspect concepts in some applications. 3. Apply AI&ML in the SDLC.	

Text Books:

- 1. Software Reuse Architecture, Process and Organization for Business Success- by Ivar Jacobson, Martin Griss, Patrick Johnson, First Edition, Pearson Education, 2000.
- 2. Software Engineering: Concepts & Practices- Ugrasen Suman, Cengage Learning Publications, Second Ed. 2022.

3. Software Engineering- Ian Somerville, Pearson Education Asia, 10th Edition, 2016.
4. Software Engineering: A Practitioner's Approach- R. S. Pressman, Tata McGraw-Hill International Editions, New York.

Reference Books:

Online Resources:

CS-6418: ADVANCED DATABASE MANAGEMENT SYSTEM

Course Outcomes (COs):

- CO1:** Build a strong foundation of query languages through relational algebra, calculus & QBE.
- CO2:** Design conceptual, logical & physical database models through ER model and normalization.
- CO3:** Develop SQL proficiency on simple & advanced features with advanced functions like Concurrency, transactions in processing, and recovery in a multiuser environment
- CO4:** Acquire necessary skills for NoSQL-based database application development.
- CO5:** Exposure to graph, spatial, and temporal databases.
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Course Contents

UNIT-I

No. of Hours: 8

Introduction: Advantages of DBMS approach, Various views of data, data independence, schema & sub-schema, primary concept of data models, database languages, transaction management, database administrator & user, data dictionary, database architectures. ER model: Basic concept, design issues, mapping constraints, keys, ER diagram, weak & strong entity-sets, specialization & generalization, aggregation, inheritance, design of ER schema, Reduction of ER Schema to tables. Domains, relation, kind of relation, Relational databases, Various types of keys: candidate, primary, alternate & foreign keys.

UNIT-II

No. of Hours: 8

Relational Algebra and SQL: The structure, relational algebra with extended operations, modification of a database, Idea of relational calculus, basic structure of SQL, set operations, Aggregate functions, Null values, Nested Sub queries, derived relations, views, Modification of database, join relation, Domain, relation & keys, DDL in SQL.

UNIT-III

No. of Hours: 8

Functional dependencies & Normalization: basic definitions, Trivial & nontrivial dependencies, closure set of dependencies & of attributes, Irreducible set of dependencies, introduction to normalization, non-loss decomposition, FD diagram, first, second and third normal forms, dependency preservation, BCNF, multivalued dependencies and fourth normal form, join dependencies and fifth normal form. Database Integrity: general idea, integrity rules, Domain rules, Attribute rules, assertions, triggers, integrity & SQL.

UNIT-IV

No. of Hours: 8

Transaction, Concurrency & Recovery: basic concept, ACID properties, transaction state, Implementation of atomicity & durability, Concurrent execution, Basic idea of serializability, Basic idea of concurrency control, basic idea of deadlock, Failure Classification, storage structure types, stable storage implementation, data access, recovery & Atomicity: log-based recovery, deferred database modification, immediate, query processing and optimization.

UNIT-V**No. of Hours: 8**

Overview of Graph & Spatial Databases & NOSQL.

Text Books:

1. Database System Concepts –Henry F. Korth, Tata McGraw-Hill, 6th Edition.
2. NoSQL Distilled- Pramod J. Sadalage, Martin Fowler, Addison-Wesley, 2013

Reference Books:

1. Fundamentals of Database Systems, Elmasri R, Navathe S, Addison Wesley, 6th Ed.
2. An Introduction to Database Systems- Bipin C. Desai
3. An Introduction to Database Systems- C.J Date
4. SQL, PL/SQL: The programming language of Oracle-Ivan Bayross

Online Resources:

1. https://onlinecourses.nptel.ac.in/noc22_cs91
2. https://onlinecourses.swayam2.ac.in/cec19_cs05

CS-6418: ADVANCED DATABASE MANAGEMENT SYSTEMS - PRACTICAL

Course Outcomes (COs):

CO1: Knowledge of SQL queries and relational algebra.

CO2: Apply normalization techniques for refining databases.

CO3: Construct triggers, procedures, functions, packages, and cursors using PL/SQL.

CO4: Construct a database for Case Study/ Real Life Problem.

CO5: Knowledge of NoSQL, Graph, and Spatial Database.

Course Contents

UNIT-I

No. of Hours: 8

1. Database Requirement Analysis

Select one real-world application such as:

- Hospital Management
- Smart City Management
- University ERP
- Digital Library

Tasks

- Gather system requirements.
- Identify entities and attributes.
- Identify relationships.
- Identify business rules.
- Prepare requirement specification document.

2. ER Model Design

Create an ER model for the selected system.

Include:

- Strong entities
- Weak entities
- Generalization
- Specialization
- Aggregation
- Cardinality
- Participation Constraints

3. ER to Relational Mapping

Convert the ER model into relational tables.

Include:

- Primary Keys
- Foreign Keys
- Composite Keys

- Candidate Keys

UNIT-II

No. of Hours: 8

1: Database Creation

Create the complete database designed in Unit I.

Perform:

1. Apply:

- Primary key
- Foreign key
- CHECK
- UNIQUE
- DEFAULT

2. Populate each major table with at least 25 records.

2. SQL Query Implementation

Implement the following:

- Selection
- Projection
- Join operations
- Nested queries
- Aggregate functions
- GROUP BY
- HAVING
- Set operations
- Views
- Derived relations
- Update
- Delete

3. Relational Algebra

Write relational algebra expressions for:

- Selection
- Projection
- Union
- Difference
- Cartesian Product
- Join
- Division

4. Advanced SQL Programming

Develop:

- Stored Procedures
- Functions
- Triggers

- Views
- Transactions

UNIT-III

No. of Hours: 8

1. Database Integrity and Triggers

Implement:

- Domain constraints
- Entity integrity
- Referential integrity
- CHECK constraints
- Triggers for:

Audit logging

Automatic stock update

Attendance calculation

Salary update

Test all constraints with valid and invalid inputs.

2. Transaction Processing

Demonstrate:

- COMMIT
- ROLLBACK
- SAVEPOINT
- Nested transactions (where supported)

Document the effect of each operation.

UNIT-IV

No. of Hours: 8

1. MongoDB CRUD Operations

Create a MongoDB database.

Perform:

- Insert
- Update
- Delete
- Find
- Sort
- Projection

Populate the collection with at least 20 documents.

2. MongoDB Aggregation Framework

Implement aggregation pipelines using:

- \$match
- \$group
- \$project
- \$sort

- \$lookup
- \$unwind

Generate analytical reports from the dataset.

UNIT-V

No. of Hours: 8

1. Graph Database using Neo4j

Design a graph database for one of the following:

- Social Network
- Research Collaboration Network
- Airline Reservation System

Implement:

- Node creation
- Relationship creation
- Ciper queries
- Shortest path analysis
- Recommendation queries

2. Spatial Database using PostGIS

Create a spatial database.

Perform:

- Creation of spatial tables
- Point and Polygon geometry
- Distance calculation
- Buffer analysis
- Spatial intersection
- Containment queries

Document all results with maps or screenshots.

Text Books:

1. NoSQL Distilled- Pramod J. Sadalage, Martin Fowler, Addison-Wesley, 2013
2. SQL, PL/SQL: The programming language of Oracle-Ivan Bayross.

Reference Books:

1. Oracle Database 10g: The Complete Reference, Kevin Loney.

Online Resources:

1. https://onlinecourses.nptel.ac.in/noc22_cs91
2. https://onlinecourses.swayam2.ac.in/cec19_cs05